Code Optimization (30 points)

1) Increment Y (5 points)

```
;Y <- Y + 1
LDI R16, 1
ADD YL, R16
LDI R16, 0
ADC YH, R16
```

```
ADIW Y, 1
```

2) Set interrupt flag and return (3 points)

```
;set interrupt flag and return
SEI
RET
```

```
RETI
```

3) Absolute value of `R16` (7 points)

Given: `R16` - 8-bit signed value

Result: `R16` - absolute value of the original value in `R16` (`R16` if `R16` >= 0 and -`R16` if `R16` < 0)

Code: can be done in 5 instructions

```
MOV R17, R16 -copy over r16
LSL R17      -get MSB
SBC R17,R17  -create mask for MSB operations
EOR R16,R17  -invert r16 if it was negative
SUB R16,R17  -add the 1 if carry was negative
```

4) Count number of set bits in `R17 | R16` (15 points)

Given: `R17|R16` - 16-bit unsigned value

Result: `R16` - count of number of set bits in `R17|R16` (value will be between 0 and 16)

Code: can be done in 26 instructions

```
;acquire count through pairs
MOV  R19, R17
MOV  R18, R16
LSR  R19          ;shift right lsb to carry
ROR  R18          ;shift right carry to msb
ANDI R19, 0x55    ;apply 01010101 mask to each
ANDI R18, 0x55    ;0101 -> 0101
SUB  R16, R18     ;1011 - 0101 = 01 | 10
                  ;01 = 1 bit in high, 10 = 2 bits in low
SUB  R17, R19     ;repeat process here as well

;sum up pair counts
MOV R19,R17
MOV R18,R16

;combined count method x = (x & 0x3333) + ((x >> 2) & 0x3333))
;I.E: 01|10 = 1 + 2 can add by them by:
;shifting top over, delete top half from both, add together
;0010(x&0x3) + 0001((x >> 2) & 0x3333) = 0011 = 3

LSR  R19
ROR  R18
LSR  R19
ROR  R18
ANDI R19, 0x33  ; Mask
ANDI R18, 0x33
ANDI R17, 0x33
ANDI R16, 0x33
ADD  R17, R19   ; Add pairs, each contains bit count
ADD  R16, R18

;repeat process with larger mask (swap instead of shifting)
ADD  R16, R17   ; add corresponding halves together
MOV  R18, R16
SWAP R18        ; swap instead of shiting
ANDI R16, 0x0F  ; same mask as last time
ANDI R18, 0x0F
ADD  R16, R18   ;updated final count
```